

Monthly Grocery Platform Services & Deployment Master Blueprint

Operational prerequisites, accounts, cloud infrastructure, APIs, and procurement roadmap for the **Monthly Grocery Platform** (Customer App, Merchant App, Web Admin Portal & Express Backend API).

Project: Monthly Grocery Ecosystem

Target Stack: React Native • Next.js 16 • Node.js • PostgreSQL

Stage: Phase 1 Launch

Architect: Innonsh Tech Team

AWS & CLOUD CORE
5 Req + 3 Rec
EC2, RDS PostgreSQL, S3, VPC, IAM, ALB

GOOGLE PLAY STORE
Android (2 Apps)
Customer & Merchant Partner Apps

APPLE APP STORE
iOS (2 Apps)
D-U-N-S, Dev Account & Test Creds

PHASE 1 SCOPE
COD & SMS First
Online PG & FCM deferred to Phase 2

1. AWS & Cloud Infrastructure Services

Hosts Express Backend REST API, PostgreSQL Database, Next.js Web Admin & Media

5 REQUIRED + 3 RECOMMENDED

SERVICE NAME	REQUIREMENT	COMPONENT	PURPOSE & WHY WE NEED IT
Amazon EC2 / Beanstalk t4g.small / t3.medium	REQUIRED	express-backend	Runs Node.js/Express API server 24/7 for order processing, catalog, merchant APIs, and auth.
Amazon RDS PostgreSQL db.t4g.micro / Supabase Pro	REQUIRED	Central Database	Stores structured business data: Customers, Merchants, Products, Delivery Slots, Monthly Baskets, Orders & Addresses with daily auto-backups.
Amazon S3 Storage Object Storage Bucket	REQUIRED	Media & Documents	Stores product photos, merchant banners, user avatars, generated PDF tax invoices, and Excel product catalogs uploaded via Web Admin.
Amazon VPC & Sec. Groups Virtual Private Cloud	REQUIRED	Network Security	Isolates PostgreSQL database in a private subnet, preventing public internet exposure while allowing secure backend communication.
AWS IAM Identity & Access Mgmt	REQUIRED	Security & Permissions	Controls programmatic API keys for S3 image uploads and least-privilege server admin credentials.
Load Balancer (ALB) Application Load Balancer	RECOMMENDED	Traffic Routing & SSL	Distributes incoming API traffic, terminates SSL/TLS, and enables zero-downtime rolling backend updates.
AWS Cert. Manager (ACM) SSL/TLS Certificates	RECOMMENDED	HTTPS Security	Provides free, auto-renewing SSL certificates for <code>api.monthlygrocery.com</code> and <code>admin.monthlygrocery.com</code> .
Route 53 / Cloudflare Domain DNS Management	CONDITIONAL	DNS Routing	Resolves domain names to AWS ALB and Vercel/Amplify Admin dashboard with DDoS mitigation.

💡 **Web Admin Hosting:** The Next.js Admin portal (`web-admin`) can be deployed seamlessly to **Vercel Pro** (\$20/mo) or AWS Amplify for automated CI/CD and SSR performance.



2. Google Maps Platform & Location APIs

Required for Customer Address Pinning, Merchant Delivery Radius & Geocoding

REQUIRED

GOOGLE API	REQUIREMENT	APP FLOW	FUNCTIONALITY
Places API (Autocomplete)	REQUIRED	Customer & Merchant Apps	Autocomplete search when customers type building, apartment, society, or street names.
Geocoding & Reverse Geocoding	REQUIRED	Map Pin Screen	Converts GPS marker coordinates (lat/lng) into structured flat/street/pincode addresses.
Maps SDK for Android & iOS	REQUIRED	Mobile Native Maps	Renders interactive vector Google Maps inside React Native apps for visual address confirmation.
Distance Matrix API	RECOMMENDED	Delivery Zone Engine	Calculates road distance between customer and assigned store to verify serviceability and delivery fees.

⚠ **Security Restriction:** Android key restricted by Package Name + SHA-1 fingerprint; iOS key restricted by Bundle Identifier (`com.monthlygrocerymobile`).



3. Google Play Store Deployment (Android)

Prerequisites to publish Customer App & Merchant Partner App on Google Play

REQUIRED

1. Google Play Console (\$25 One-time)

Developer account with verified legal identity proof (Govt ID + Business Address proof).

2. 20-Tester Closed Testing (14 Days)

Google mandate for accounts created post-Nov 2023: 20 opted-in testers active for 14 continuous days.

3. Release Keystore & App Signing

Secure production `.keystore` file, alias, passwords, and Google Play App Signing SHA-256 fingerprint.

4. Store Graphic Assets

512x512 PNG App Icon, 1024x500 Feature Graphic banner, and minimum 4 phone screenshots per app.

5. Data Safety Form

Declarations for phone number (auth), precise location (delivery), and device identifiers in Play Console.

6. Account Deletion Compliance

In-app account deletion workflow + Public web account deletion URL (already implemented in project).



4. Apple App Store Deployment (iOS)

Prerequisites to publish iOS Customer & Partner apps on App Store Connect

REQUIRED

1. Apple Developer Program (\$99/yr)

Organization or Individual enrollment with valid international payment method.

2. D-U-N-S Number (Organization)

Free 9-digit business legal entity identifier from Dun & Bradstreet (required for company accounts).

3. Demo Review Credentials

Working demo mobile number + bypass OTP provided in App Review Information for Apple test team.

4. App Privacy Nutrition Labels

Disclosures for Location (order delivery), Contact Info (phone OTP), and usage telemetry.

5. iOS Display Screenshot Sets

6.7-inch (iPhone 16 Pro Max) & 6.5-inch display screenshots capturing home, cart, slot & basket flows.

6. iOS Distribution Certificate

Production `.p12` Distribution Certificate and App Store provisioning profiles configured in Xcode.



5. SMS OTP & DLT Compliance (India Telecom)

Mandatory regulatory setup to deliver customer login & order OTPs

REQUIRED

SERVICE / ITEM	PROVIDER OPTIONS	REQUIREMENT	DETAILS & SETUP STEPS
DLT Entity Registration TRAI Mandate	Jio DLT / Airtel / Vilpower	MANDATORY	Register company entity with PAN, GST, and Certificate of Incorporation (~₹5,000 + GST).
SMS Sender ID (Header) e.g. MNTHGR / INNSH	DLT Telecom Portal	MANDATORY	6-character registered header approved for transactional OTP SMS delivery.
SMS Gateway API Fast Delivery Gateway	Fast2SMS / MSG91 / Twilio	REQUIRED	API Key plugged into <code>express-backend/src/routes/auth.ts</code> to send 6-digit OTPs in <3s.



6. Payment Gateway Integration

Online Payment Processing (UPI, Cards, NetBanking, AutoPay)

NOT IN SCOPE FOR PHASE 1 (DEFERRED TO PHASE 2)



Phase 1 Operating Model (Cash on Delivery / Direct Store Settlement):

For the **Phase 1 Initial Launch**, online payment aggregator integration (Razorpay / PhonePe / Cashfree) is **NOT in scope**. Orders will be processed using **Cash on Delivery (COD) / Pay on Delivery / Direct Store Settlement**. No payment gateway business KYC, merchant aggregator accounts, or webhook secrets are required for Phase 1 deployment.

PHASE 2 TARGET SERVICE	SUPPORTED MODES	PHASE 1 STATUS	PHASE 2 ROADMAP
Razorpay / PhonePe / Cashfree Payment Aggregator	UPI Intent, Cards, NetBanking, AutoPay	NOT IN PHASE 1	Will be activated in Phase 2 with merchant KYC (GST, Current Account, Cancelled Cheque) and automated webhook callbacks.



7. Push Notifications & Background Alerts

Mobile Background Push Notifications for Order Updates & Basket Reminders

NOT IN SCOPE FOR PHASE 1 (DEFERRED TO PHASE 2)



Phase 1 Notification Strategy:

For the **Phase 1 Initial Launch**, mobile push notification infrastructure (FCM & APNs) is **NOT in scope**. Customers and merchants will track orders via **In-App Live Order Status Tracking** and **SMS Transactional Alerts**. Firebase Cloud Messaging (FCM) v1 service keys and APNs Auth Keys (.p8) will be configured in **Phase 2**.



8. Transactional Email & Customer Support

PDF Invoice Delivery, Merchant Statements & Help Desk

RECOMMENDED

SERVICE TYPE	RECOMMENDED PROVIDER	REQUIREMENT	KEY DELIVERABLES
Amazon SES / Resend Transactional SMTP	Amazon SES or Resend.com	RECOMMENDED	Automated delivery of Order Summary PDF Invoices and Monthly Merchant Settlement reports. Requires SPF, DKIM, DMARC DNS setup.
Help Desk / WhatsApp Support Customer Helpline	WhatsApp Business / Email	OPTIONAL	Support helpline button inside mobile app pointing to company support email or WhatsApp helpline.



9. Mandatory Legal Policies & Store URLs

Must be published on live public web URLs before submitting apps to App Store & Play Store

REQUIRED FOR STORE APPROVAL

■ 1. Privacy Policy Webpage

Discloses data collection (phone number, GPS location for grocery delivery, device telemetry, analytics).

■ 2. Terms & Conditions (EULA)

Defines minimum order thresholds (₹3,000 limit), recurring monthly basket rules, and delivery commitments.

■ 3. Return, Refund & Cancellation

Specifies rules for damaged perishable items, wrong product deliveries, and refund turnaround timelines.

■ 4. Shipping & Delivery Policy

Details operational delivery slot windows (Morning 7–10 AM, Evening 5–9 PM) and serviceable areas.

■ 5. Public Web Account Deletion Request URL (Mandatory by Apple & Google)

Live webpage where registered users can request account & data deletion without installing the mobile app.



10. Estimated Budget & Step-by-Step Procurement Roadmap

Actionable roadmap for the management and procurement team

ACTION PLAN

ONE-TIME REGISTRATIONS

₹15,000 – ₹20,000

Play Console + Apple Dev + DLT + Domain

AWS CLOUD HOSTING

₹3,500 – ₹8,000 /mo

EC2, RDS PostgreSQL, S3 (Free tier applies)

GOOGLE MAPS PLATFORM

₹0 – ₹2,000 /mo

Google provides \$200 free monthly credit

PHASE 1 PAYMENT MODEL

COD (₹0 Gateway Fee)

Online PG deferred to Phase 2

1 Create Master Cloud Accounts

Register **AWS Root Account** (linked to business card) & **Google Cloud Console Account** (for Maps APIs). Register domain (monthlygrocery.in).

2 Apply for Developer & DLT Accounts

Obtain D-U-N-S number, register **Google Play Console** (\$25) and **Apple Developer** (\$99/yr). Submit DLT entity registration on Jio/Airtel DLT portal for SMS OTPs.

3 Publish Legal Compliance Pages

Host Privacy Policy, Terms & Conditions, Return/Refund Policy, and Account Deletion Request webpage on the live domain.

4 Hand Over Keys to Deployment Team

Provide AWS IAM credentials, Google Maps API Keys, DLT Sender ID, and Apple/Play store access to the deployment engineering team to build release bundles.